Prediction of Life Evaluation Using Machine Learning

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Abstract

This study develops machine learning models to predict country-level life evaluation scores using a combination of data from the World Happiness Report, World Bank Group, and Human Development Index. A dataset of 1,567 country-year observations with 506 features was constructed and used to train multiple models. Feature selection using LASSO reduced the feature set to 64 while maintaining predictive performance. The best-performing model achieved a test RMSE of 0.277 and an R^2 of 0.946. Results indicate that incorporating external variables can improve predictive performance when compared to approaches relying solely on World Happiness Report variables.

1. Introduction

Understanding the factors that drive happiness has become an increasingly important area of research in economics and public policy, particularly as traditional measures of growth such as GDP fail to fully capture overall well-being. The World Happiness Report addresses this gap by providing life evaluation as a standardized measure of happiness, based on survey responses in which individuals rate their lives on a scale of one to ten. While this measure provides a useful benchmark to compare well-being across countries, the explanatory variables included within the report are relatively limited in scope.

To address this limitation, this study integrates data from multiple sources, including the World Happiness Report, the World Bank Group, and the United Nations Human Development Index. These additional datasets provide a broader and more detailed set of socioeconomic and developmental indicators, such as access to basic resources, infrastructure, and overall economic conditions. By combining these sources, the analysis moves beyond the broader indicators provided by the WHR and incorporates more concrete measures that may influence life evaluation.

The primary objective of this project is to develop predictive models that can accurately predict country-level life evaluation scores, while also identifying the key factors that contribute to these outcomes. Due to the complexity and dimensionality of the data, machine learning techniques are used to capture nonlinear relationships and interactions among variables that traditional methods may overlook. In addition to predictive performance, emphasis was placed on model interpretability through feature selection, allowing for a more focused analysis of the variables that most significantly influence happiness.

Ultimately, this study aims to demonstrate that incorporating a richer set of external data sources can improve predictive performance and provide deeper insight into

the drivers of global well-being. These findings contribute to a more comprehensive understanding of happiness and offer potential value for policymakers seeking to improve quality of life across different countries and regions.

2. Literature Review

The study of human happiness and quality of life has gained increasing attention with the availability of large-scale datasets such as the World Happiness Report. This report quantifies life evaluation through an administered survey and incorporates a limited set of variables including log GDP per capita, healthy life expectancy, and freedom from corruption. While these variables provide a useful foundation, they may not fully capture the multidimensional nature of happiness.

Recent research has explored the application of machine learning techniques to improve the prediction of happiness and quality of life indicators. Jannani et al. conducted a study comparing the effectiveness of multiple machine learning and deep learning models on WHR data. Their results showed that regression-based approaches achieved the best performance, with their highest recorded R^2 value being approximately 0.895. This suggests that when working with standard WHR features, linear models are sufficient to capture the variance present in life-evaluation ratings.

Similarly, Bloch et al. applied a range of machine learning models; Support Vector Regression, Decision Trees, Random Forest, and CatBoost, to predict life-evaluation scores using WHR variables. Their results showed their best performing model was an SVR achieving an R^2 value of approximately 0.827.

More recent work has begun to explore hybrid approaches that combine unsupervised and supervised learning techniques. Yang and Xie proposed a two-stage method that integrates K-Means clustering with machine learning models like Random Forest and XGBoost. In this approach, countries are grouped into clusters, and these cluster assignments are then used as additional features in the predictive models. This method resulted in an overall improvement in predictive performance, with approximately a 12% increase in R^2 compared to models that did not incorporate clustering information. Their results further confirmed that GDP and social support are the most influential predictors of happiness.

Across these studies, a pattern emerges: most studies rely exclusively on World Happiness Report variables, resulting in models that capture only high-level relationships and often plateau in performance. The WHR itself explains that the features included in their report can explain just above ¾ of the variation present. This study extends on prior work by integrating data from multiple sources, including the World Bank Group and the Human Development Index. In contrast to prior studies reporting R^2 values between 0.62 and 0.89, this study evaluates whether expanding the feature space can improve predictive performance.

3. Methodology

3.1 Data Sources

This study combines data from three primary sources: the World Happiness Report (WHR), the World Bank Group (WBG), and the United Nations Human Development Index (HDI). The dependent variable, *Life Evaluation*, is taken from the WHR and represents country-level happiness based on responses to a single survey question:

Please imagine a ladder with steps numbered from 0 at the bottom to 10 at the top. The top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step of the ladder would you say you personally feel you stand at this time?

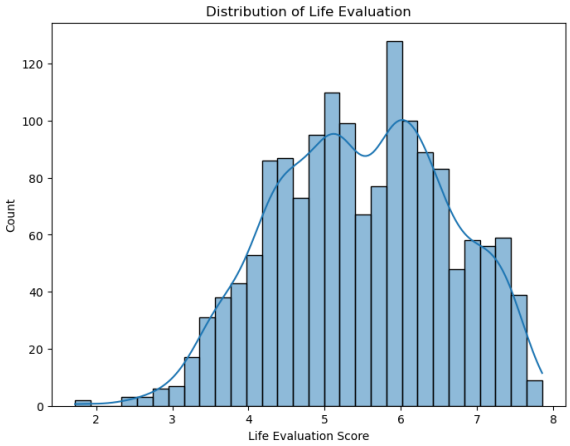
The WHR has been collecting this data since 2005, however data is only available dating back to 2011. They collect data from over 140 participating countries; however, some countries are not accounted for in every year of the poll due to difficulties in administering the survey. Ultimately, the final dataset obtained from the WHR contains 1,969 entries each containing a country and the associated year that the survey was administered.

The data coming from the WBG and HDI functions as supplementary data to that of the WHR. While the WHR does contain some data relevant to the task at hand, the supplemental data provides a more detailed compilation of factors that might be relevant to global happiness. Factors like 'Healthy Life Expectancy' which was present in the WHR can now be represented by more robust features such as access to clean water and government health expenditure as a percentage of total GDP. The HDI provided 26 variables, while the WBG provided 587. After removing duplicate and sparsely populated variables, the final feature set was comprised of 506 variables.

3.2 Data Preprocessing

Given that the data originated from multiple sources, preprocessing was primarily focused on ensuring consistency and alignment across datasets. The first step involved standardizing country names to enable accurate joins between the WHR, WBG, and HDI datasets. Variations in naming conventions (e.g., abbreviations, alternate spellings) were resolved to create a unified country identifier. Additionally, datasets were reshaped to ensure a consistent structure of country-year observations. Countries that did not contain data from the WBG or HDI were excluded from the final dataset.

Following integration, the dataset contained a substantial amount of missing data, particularly within the World Bank variables. To address this, columns with excessive missing values or limited relevance were removed. For the remaining variables, missing values were imputed using a nearest-year approach within each country, where available observations from adjacent years were used to fill gaps. In cases where no country-specific data was available, global averages were used as a fallback to preserve the feature while minimizing bias. All imputation steps were performed prior to model training. After preprocessing, the final dataset consisted of 1,567 country-year observations with 506 candidate features.



3.3 Modeling

To predict life evaluation scores, a range of supervised machine learning models were developed and evaluated. In total, nine individual models were trained, including both linear and tree-based approaches. In addition to the individual models, an ensemble model was constructed using a stacking approach. The top-performing models, Gradient Boosting and XGBoost, were combined to generate final predictions.

To address the high dimensionality of the dataset, all models were trained using two separate features sets: the full set of 506 variables and a reduced subset obtained through LASSO-based feature selection. Feature selection using LASSO was performed using training data only to avoid information leakage. Features identified through LASSO-based feature selection were further examined and, where appropriate, transformed to better capture their relationship with the dependent variable. The reduced feature set contained 64 variables and was intended to improve the interpretability of the models while maintaining strong predictive performance. This approach allowed for a direct comparison between model performance when using a comprehensive set and a more focused set of features.

Model performance was evaluated using Root Mean Squared Error (RMSE) and R^2 on a held-out test set. The test set contained all entries from the year 2023, the most recent year of data available, and represented 128 different countries. This time-based split was chosen to better reflect real-world predictive scenarios and to evaluate model generalization on unseen future data. The models evaluated included Multiple Linear Regression, Ridge, LASSO, Support Vector Regression, Decision Trees, Random Forest, AdaBoost, Gradient Boosting, and XGBoost.

4. Results

4.1 Model Performance

The results showcase a clear performance gap between model types, with tree-based ensemble methods consistently outperforming linear and simple approaches. When trained on the full feature set, XGBoost achieved the strongest test set performance (RMSE = 0.279, R^2 = 0.945), followed by Random Forest (RMSE = 0.319, R^2 = 0.928) and Gradient Boosting (RMSE = 0.326, R^2 = 0.924). In contrast, linear models such as Ridge (RMSE = 0.45, R^2 = 0.856) and LASSO (RMSE = 0.451, R^2 = 0.855) showed notably weaker performance. This suggests that the relationship between the predictors and life evaluation is inherently nonlinear and not captured as well by linear models.

A comparison between training and test performance highlights differences in

Model Results ~ Full Feature Set				
Model	**Train RMSE**	**Train R^2**	**Test RMSE**	**Test R^2**
XGBoost	0.123	0.988	0.279	0.945
Random Forest	0.114	0.99	0.319	0.928
GradientBoost	0.193	0.971	0.326	0.924
SVR	0.204	0.967	0.38	0.897
AdaBoost	0.351	0.903	0.406	0.883
Decision Tree	0.182	0.974	0.439	0.863
Ridge	0.381	0.886	0.45	0.856
LASSO	0.397	0.876	0.451	0.855
MultiLinear	0.434	0.816	0.571	0.768

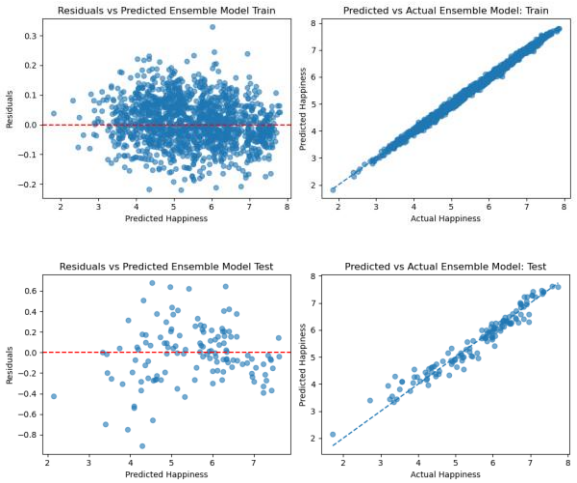
Model Results ~ Reduced Feature Set				
Model	**Train RMSE**	**Train R^2**	**Test RMSE**	**Test R^2**
Ensemble	0.076	0.995	0.277	0.946
GradientBoost	0.073	0.996	0.282	0.944
XGBoost	0.08	0.995	0.284	0.943
Random Forest	0.116	0.989	0.327	0.924
SVR	0.197	0.97	0.398	0.887
Decision Tree	0.126	0.988	0.429	0.869
LASSO	0.374	0.89	0.433	0.867
Ridge	0.406	0.87	0.45	0.856
AdaBoost	0.389	0.881	0.465	0.846
MultiLinear	0.417	0.863	0.486	0.832

model generalization. While several models achieved extremely high training R^2 values, this did not always translate to equivalent test performance. Random Forest, for example, showcased a noticeable drop from a training R^2 of 0.99 to 0.928 on the test set, indicating some degree of overfitting. XGBoost, maintained a more consistent performance between training and testing, suggesting that it is better at generalizing to unseen data.

The introduction of feature selection produced slight improvements in model behavior. Reducing the feature space from 506 to 64 resulted in minimal loss of predictive accuracy and, in some cases, notable gains. Gradient Boosting benefited significantly from this reduction, with its test RMSE decreasing from 0.326 to 0.282 and R^2 increasing from 0.924 to 0.944. This indicates that the full feature set likely contained redundant or noisy variables that hindered model performance. The reduced feature set therefore improved both interpretability and generalization by focusing on the most informative features.

Building on these results, the ensemble model combining Gradient Boosting and XGBoost achieved the best overall performance, with a test RMSE of 0.277 and R^2 of 0.946. Although the improvement over the best individual model was minor, the ensemble model consistently produced the lowest prediction error. This suggests that the models capture slightly different patterns

within the data, and that combining them allows for more accurate predictions.



4.2 Feature Importance

To gain a better understanding of the drivers of life evaluation, feature importance scores from the ensemble model were collected and examined. The results reveal an uneven distribution of importance across variables. The most influential variable by a large margin was Gross National Income (GNI) per capita. This indicates that economic prosperity is a key predictor of life evaluation. This finding aligns with existing research and common-sense knowledge that income and economic stability are key contributors to a person's well-being.

Beyond income, several features related to health and access to basic resources emerged as important predictors. Variables such as domestic government healthcare expenditure, access to electricity, access to clean fuel for cooking, and coverage for reproductive health all contributed meaningfully to the model's performance. These results suggest that quality of life factors, particularly those related to infrastructure and public health, are also strongly associated with higher life evaluation scores. Additionally, other macroeconomic indicators such as the consumer price index and trade-related variables (merchandise imports) also appeared among the top predictors.

Overall, the feature importance analysis shows that while economic output is the strongest single predictor of life evaluation, a combination of health, infrastructure, and macroeconomic stability variables collectively contribute to explaining differences in happiness. This reinforces the importance of considering multiple dimensions of development when modeling well-being. A full table containing the feature importance scores can be found at the end of this report.

5. Conclusion

The goal of this study was to develop predictive models for country-level life evaluation scores, while gaining a deeper understanding of the factors that drive global happiness. By integrating data from the World Happiness Report, the World Bank Group, and the Human Development Index, a comprehensive dataset was constructed that extended beyond the limited scope of traditional happiness analyses.

The results demonstrate that machine learning models, particularly tree-based methods, are highly effective in predicting life evaluation scores. Models such as XGBoost and Gradient Boosting achieved strong predictive performance, indicating that a large amount of the variation in country-level happiness can be explained by observable socioeconomic factors.

The key contribution of this study is the demonstration that incorporating external datasets can potentially enhance model performance compared to approaches that rely solely on World Happiness Report variables. The inclusion of detailed indicators related to economic conditions, public health, and infrastructure provided a more complete representation of the factors influencing life evaluation. Feature importance analysis revealed that while income is the dominant predictor, other factors such as access to healthcare, electricity, and basic resources play a critical role as well.

Future work could explore clustering countries into more homogeneous groups, such as income-based or regional clusters, to account for differences in how happiness is determined. Additionally, the application of deep learning models may provide further improvements in predictive accuracy. As data quality and availability continues to improve over time, revisiting this analysis with more complete datasets may provide even deeper insight into the drivers of global well-being.

Feature Importance Table	
Feature	**Importance**
GNI_per_capita_Atlas_method_current_US_NY_GNP_PCAP_CD_	0.34449517
Domestic_general_government_health_expenditure_of_GDP_SH_XPD_GHED_GD_ZS_	0.01803168
Access_to_electricity_of_population_EG_ELC_ACCS_ZS_	0.01669668
Consumer_price_index_2010_100_FP_CPI_TOTL_	0.01366083
UHC_service_coverage_sub_index_on_reproductive_maternal_newborn_and_child_health_SH_UHC_SCI_RMNCH_	0.01271312
Access_to_clean_fuels_and_technologies_for_cooking_urban_of_urban_population_EG_CFT_ACCS_UR_ZS_	0.01182272
Log_GNI_per_capita_PPP_current_international_NY_GNP_PCAP_PP_CD_	0.01017808
Employment_to_population_ratio_ages_15_24_male_modeled_ILO_estimate_SL_EMP_1524_SP_MA_ZS_	0.0089227
Log_Merchandise_imports_from_low_and_middle_income_economies_in_Latin_America_the_Caribbean_of_total_merchandise_imports_TM_VAL_MRCH_R3_ZS_	0.00874266
External_health_expenditure_of_current_health_expenditure_SH_XPD_EHEX_CH_ZS_	0.00814103
Import_volume_index_2015_100_TM_QTY_MRCH_XD_WD_	0.00756839
Log_Agriculture_forestry_and_fishing_value_added_per_worker_constant_2015_US_NV_AGR_EMPL_KD_	0.00754962
External_health_expenditure_per_capita_PPP_current_international_SH_XPD_EHEX_PP_CD_	0.00725238
Merchandise_exports_to_economies_in_the_Arab_World_of_total_merchandise_exports_TX_VAL_MRCH_AL_ZS_	0.00684713
Prevalence_of_anemia_among_non_pregnant_women_of_women_ages_15_49_SH_ANM_NPRG_ZS_	0.00606802
Unemployment_female_of_female_labor_force_modeled_ILO_estimate_SL_UEM_TOTL_FE_ZS_	0.0058283
UHC_service_coverage_sub_index_on_non_communicable_diseases_SH_UHC_SCI_NCD_	0.00573996
Merchandise_exports_to_low_and_middle_income_economies_in_Sub_Saharan_Africa_of_total_merchandise_exports_TX_VAL_MRCH_R6_ZS_	0.00519657
Merchandise_exports_to_low_and_middle_income_economies_in_Latin_America_the_Caribbean_of_total_merchandise_exports_TX_VAL_MRCH_R3_ZS_	0.00504723
Mobile_cellular_subscriptions_per_100_people_IT_CEL_SETS_P2_	0.00500455
Contributing_family_workers_female_of_female_employment_modeled_ILO_estimate_SL_FAM_WORK_FE_ZS_	0.00497854
Share_of_youth_not_in_education_employment_or_training_male_of_male_youth_population_modeled_ILO_estimate_SL_UEM_NEET_MA_ME_ZS_	0.0044516
Population_in_the_largest_city_of_urban_population_EN_URB_LCTY_UR_ZS_	0.00441927
Unemployment_male_of_male_labor_force_modeled_ILO_estimate_SL_UEM_TOTL_MA_ZS_	0.00426396
Domestic_general_government_health_expenditure_of_general_government_expenditure_SH_XPD_GHED_GE_ZS_	0.004004
Merchandise_imports_from_economies_in_the_Arab_World_of_total_merchandise_imports_TM_VAL_MRCH_AL_ZS_	0.00393446
Capture_fisheries_production_metric_tons_ER_FSH_CAPT_MT_	0.00347507
Log_Price_level_ratio_of_PPP_conversion_factor_GDP_to_market_exchange_rate_PA_NUS_PPPC_RF_	0.0034129
Domestic_private_health_expenditure_per_capita_current_US_SH_XPD_PVTD_PC_CD_	0.00311802
Share_of_seats_in_parliament_female_held_by_women_	0.00308567
Inflation_consumer_prices_annual_FP_CPI_TOTL_ZG_	0.00278711
Employment_in_industry_female_of_female_employment_modeled_ILO_estimate_SL_IND_EMPL_FE_ZS_	0.00276648
Rural_population_of_total_population_SP_RUR_TOTL_ZS_	0.00262513
Permanent_cropland_of_land_area_AG_LND_CROP_ZS_	0.00253175
Death_rate_crude_per_1_000_people_SP_DYN_CDRT_IN_	0.00251499
Preprimary_education_duration_years_SE_PRE_DURS_	0.00249626
Population_ages_65_and_above_female_SP_POP_65UP_FE_IN_	0.00245864
Prevalence_of_HIV_total_of_population_ages_15_49_SH_DYN_AIDS_ZS_	0.00232905
Merchandise_exports_to_low_and_middle_income_economies_in_East_Asia_Pacific_of_total_merchandise_exports_TX_VAL_MRCH_R1_ZS_	0.00229846
Merchandise_imports_from_low_and_middle_income_economies_in_Middle_East_North_Africa_of_total_merchandise_imports_TM_VAL_MRCH_R4_ZS_	0.00222989
People_using_at_least_basic_drinking_water_services_rural_of_rural_population_SH_H2O_BASW_RU_ZS_	0.00216622
Control_of_Corruption_Standard_Error_CC_STD_ERR_	0.0021477
Domestic_general_government_health_expenditure_of_current_health_expenditure_SH_XPD_GHED_CH_ZS_	0.00213924
Number_of_under_five_deaths_SH_DTH_MORT_	0.00198267
Asylum_seekers_by_country_or_territory_of_asylum_SM_POP_ASYS_EA_	0.0018848
Log_GNI_per_capita_PPP_constant_2021_international_NY_GNP_PCAP_PP_KD_	0.0018652
Immunization_HepB3_of_one_year_old_children_SH_IMM_HEPB_	0.00184311
Tuberculosis_treatment_success_rate_of_new_cases_SH_TBS_CURE_ZS_	0.00172031
Merchandise_imports_from_high_income_economies_of_total_merchandise_imports_TM_VAL_MRCH_HI_ZS_	0.00160281
Urban_population_of_total_population_SP_URB_TOTL_IN_ZS_	0.00150962
Export_unit_value_index_2015_100_TX_UVI_MRCH_XD_WD_	0.0014889
Tuberculosis_case_detection_rate_all_forms_SH_TBS_DTEC_ZS_	0.00146487
Population_ages_75_79_female_of_female_population_SP_POP_7579_FE_5Y_	0.00136646
Fertilizer_consumption_kilograms_per_hectare_of_arable_land_AG_CON_FERT_ZS_	0.00125603
Employers_female_of_female_employment_modeled_ILO_estimate_SL_EMP_MPYR_FE_ZS_	0.00117118
Lower_secondary_school_starting_age_years_SE_SEC_AGES_	0.00104859
Secure_Internet_servers_per_1_million_people_IT_NET_SECR_P6_	0.00092096
Arable_land_of_land_area_AG_LND_ARBL_ZS_	0.00087851
Taxes_less_subsidies_on_products_constant_LCU_NY_TAX_NIND_KN_	0.00086623
Manufacturing_value_added_annual_growth_NV_IND_MANF_KD_ZG_	0.00084367
Compulsory_education_duration_years_SE_COM_DURS_	0.00081495
Asylum_seekers_by_country_or_territory_of_origin_SM_POP_ASYS_EO_	0.00055601
Exports_of_goods_and_services_annual_growth_NE_EXP_GNFS_KD_ZG_	0.00040535
Carbon_dioxide_CO2_net_fluxes_from_LULUCF_Deforestation_Mt_CO2e_EN_GHG_CO2_LU_DF_MT_CE_AR5_	9.35E-05

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